Finite-coupling spectral asymptotics for a harmonic delta-comb

Anonymous Authors

Unnumbered research manuscript draft
5 September 2026

Abstract

We prove a two-term counting law for the half-line Dirichlet Laplacian with equal positive delta interactions at the harmonic positions $x_n = \pi \sum_{j=1}^{n} j^{-1}$. For each fixed finite coupling $\kappa > 0$, the inclusive counting function satisfies

$$N_\kappa(k^2) = 2k \log k + \big[\log(4\pi/\kappa) + \gamma - 2\big] k + O_\kappa(\log k),$$

where γ is Euler's constant. Replacing the finite interactions by Dirichlet walls changes the leading high-energy coefficient, so the fixed-coupling count requires a different comparison. We instead compare endpoint sampling with a cell-average potential, identify the exact exponentially weighted form domain, and prove a parameter-uniform counting estimate for exponential-potential comparators using modified Bessel functions. An energy-dependent form comparison transfers that estimate to the delta-comb with logarithmic error. We derive the heat-trace expansion, meromorphic continuation of the source spectral zeta function to $\operatorname{Re} s > 0$, and the sharp Schatten threshold $p > 1/2$ for the inverse. As the coupling tends to infinity, compact domination upgrades the classical strong-resolvent Dirichlet limit to norm-resolvent convergence. Inclusive counts converge even at endpoint eigenvalues, yet the Dirichlet endpoint has leading coefficient one rather than two. Thus high energy and infinite coupling do not commute at the level of the leading counting coefficient.

1 Introduction

Equal repulsive point interactions can confine a particle on a half-line when their spacing decreases sufficiently fast. The harmonic chain, whose successive gaps are π/n, is a concrete example: its spectrum is discrete for every positive coupling even though the total length is infinite. Its infinite-coupling limit is particularly explicit. Dirichlet conditions at every interaction split the chain into intervals, and the multiplicity of the eigenvalue j^2 becomes the divisor count $d(j)$. The question here is how that endpoint description compares with the high-energy spectrum at a *fixed finite* coupling.

Write

$$h_0 = 0, \qquad h_n = \sum_{j=1}^{n} \frac{1}{j}, \qquad x_n = \pi h_n.$$

Let H_κ denote the self-adjoint realization of $-d^2/dx^2 + \kappa \sum_{n\geq 1} \delta_{x_n}$ on $(0,\infty)$ with a Dirichlet condition at zero. The precise closed-form realization is given in Section 2. Our main asymptotic is

$$N_\kappa(k^2) = 2k \log k + C_\kappa k + O_\kappa(\log k), \qquad C_\kappa = \log(4\pi/\kappa) + \gamma - 2, \tag{1.1}$$

where N_κ counts eigenvalues inclusively and κ is fixed in $(0, \infty)$. In contrast, the Dirichlet endpoint obeys

$$N_\infty(k^2) = k \log k + (2\gamma - 1)k + O(\sqrt{k}). \tag{1.2}$$

The change from two to one in the leading coefficient is not inconsistent with norm-resolvent convergence: the latter controls each fixed spectral window, not a window tending to infinity uniformly in the coupling.

Classical ownership and the question addressed. Egger né Endres and Steiner introduced this one-parameter chain, proved compactness and strict positivity for positive coupling, and established its strong-resolvent Dirichlet limit [4, Section 2]. They also developed its Dirichlet divisor spectrum and trace/zeta theory. Those results are inputs and context, not contributions claimed here. The final paragraph of their Section 2 defers the finite-coupling high-energy asymptotics; this describes the scope of that paper, not a certificate that the question remains open today. Egger's dissertation explicitly places that infinite-chain work outside its scope and then treats graphs with finitely many edges [3, Introduction and Chapter 2].

Later spectral-comparison work revisits the same infinite-volume chain. The modified local Weyl law in the arXiv version of Bifulco and Kerner [2, Section 5, Theorem 11] takes infinite coupling. Sections 6.3.3 and 6.3.5 of Bifulco [1] retain that endpoint, including when adding a bounded potential. These statements concern a different limiting regime from (1.1). The proposed increment of this manuscript is the explicit finite-coupling two-term count with controlled logarithmic error. The cited passages do not contain that formula; our bounded literature check does not establish global priority.

Mechanism and results. The density of vertices on the nth cell is n/π. Since $h_n - \log n \to \gamma$, averaging the point interactions suggests the potential $\kappa e^{-\gamma} e^{x/\pi}/\pi$. The useful fact is stronger than this heuristic: the error in replacing the sampled quadratic form by its cell average is bounded by $2\kappa \|f\|_2 \|f'\|_2$, with no inverse-gap loss. The cell-average potential in turn differs from the exponential by a bounded additive function. A uniform Bessel phase estimate then permits a comparison parameter of size $1/k$, giving (1.1).

The argument has three consequences that clarify the finite-coupling problem. First, it identifies the exact common form domain for all $\kappa > 0$, so the comparison is between closed operators. Second, it yields the heat expansion and zeta principal part, with continuation only to $\operatorname{Re} s > 0$, together with the sharp threshold $H_\kappa^{-1} \in \mathcal{S}_p$ if and only if $p > 1/2$. Third, norm-resolvent convergence and monotonicity settle the inclusive endpoint thresholds and prove the unequal iterated limits of the leading coefficient. Finite calculations are included as checks of constants and discretizations, not as evidence replacing any of these proofs.

Sections 2–4 prove the finite-coupling claims. Section 5 treats the singular endpoint, Section 6 reports the numerical checks, and Section 7 records the limits of the conclusions.

2 Form domain and spectral problem

Let $\mathcal{H} = L^2(0, \infty; \mathbb{C})$ and $I_n = (x_{n-1}, x_n)$, so $\ell_n = |I_n| = \pi/n$. Here $H_0^1(0, \infty)$ is the Sobolev space of square-integrable functions with square-integrable weak derivative and zero trace at zero. No trace is imposed at infinity. Every such function has a continuous representative, which is used at the vertices. For fixed $0<\kappa<\infty$, consider

$$\mathcal{D}_\delta=\left\{f\in H_0^1(0,\infty):\sum_{n\geq 1}|f(x_n)|^2<\infty\right\},$$
$$q_\kappa[f]=\|f'\|_2^2+\kappa\sum_{n\geq 1}|f(x_n)|^2. \tag{2.1}$$

Its associated operator will be denoted by H_κ. The source frequency is $k=\sqrt{E}$, and the quantum evolution is e^{-itH_κ}.

Theorem 2.1 (Finite positive coupling)**.** *For each fixed $\kappa\in(0,\infty)$ the following hold.*

1. *The form (2.1) is densely defined, nonnegative and closed, with domain*

$$\mathcal{D}_\delta=\mathcal{D}:=H_0^1(0,\infty)\cap L^2((0,\infty),e^{x/\pi}\,dx). \tag{2.2}$$

The operator H_κ has compact resolvent and a complete orthonormal eigenbasis with simple eigenvalues $0<E_1(\kappa)<E_2(\kappa)<\cdots\to\infty$.

2. *With $N_\kappa(E)=\#\{j:E_j(\kappa)\leq E\}$,*

$$N_\kappa(k^2)=2k\log k+C_\kappa k+O_\kappa(\log k),\qquad C_\kappa=\log(4\pi/\kappa)+\gamma-2. \tag{2.3}$$

The estimate holds for all sufficiently large real k, including eigenvalue thresholds. Its constants need not be uniform in κ.

3. *As $t\downarrow 0$,*

$$\operatorname{Tr}(e^{-tH_\kappa})=\frac{\sqrt{\pi}}{2\sqrt{t}}\log\frac{\pi}{\kappa t}+O_\kappa(\log(1/t)). \tag{2.4}$$

4. *The source zeta function $Z_\kappa(s)=\sum_j E_j(\kappa)^{-s}$, initially absolutely convergent for $\operatorname{Re}s>1/2$, has a meromorphic continuation to $\operatorname{Re}s>0$. Its only pole there is at $s=1/2$, with principal part*

$$\frac{1/2}{(s-1/2)^2}+\frac{1+C_\kappa/2}{s-1/2}. \tag{2.5}$$

5. *For every real $p>0$, $H_\kappa^{-1}\in\mathcal{S}_p$ if and only if $p>1/2$. The same threshold holds for each fixed resolvent $(H_\kappa-z)^{-1}$, $z\in\rho(H_\kappa)$. In particular the ordinary Fredholm determinant $\det(I-zH_\kappa^{-1})$ exists.*

The symbol $\mathcal{S}_p$ means summability of the pth powers of singular values, also when $0<p<1$. The first part of the theorem is established below. Its compactness conclusion is classical for this model [4]; the proof supplies the weighted domain and quantitative comparison used in the later parts.

2.1 Sampling and exponential comparison

Define

$$V_\kappa(x)=\frac{\kappa n}{\pi}\quad(x\in I_n),\qquad C=\frac{\kappa e^{-\gamma}}{\pi},\qquad B=\frac{\kappa}{\pi}. \tag{2.6}$$

Lemma 2.2 (Sampling before infinite summation)**.** *For every $f\in H_0^1(0,\infty)$ and integer $N\geq 1$,*

$$\left|\kappa\sum_{n=1}^{N}|f(x_n)|^2-\int_0^{x_N}V_\kappa(x)|f(x)|^2\,dx\right|\leq 2\kappa\left\|f\right\|_2\left\|f'\right\|_2. \tag{2.7}$$

Consequently,

$$\mathcal{D}_\delta = H_0^1(0,\infty) \cap L^2(V_\kappa\, dx),$$

$$\left| \kappa \sum_{n\geq 1} |f(x_n)|^2 - \int_0^\infty V_\kappa |f|^2 \right| \leq 2\kappa \, \|f\|_2 \, \|f'\|_2 \quad (f \in \mathcal{D}_\delta). \tag{2.8}$$

Proof. Absolute continuity of the Sobolev representative on one cell gives

$$||f(x_n)|^2 - |f(x)|^2| \leq 2 \int_x^{x_n} |f(y) f'(y)|\, dy.$$

Averaging over $x \in I_n$ and interchanging the nonnegative integrals yield

$$\begin{aligned}\left| |f(x_n)|^2 - \frac{1}{\ell_n} \int_{I_n} |f|^2 \right| &\leq \frac{2}{\ell_n} \int_{I_n} (y - x_{n-1}) |f f'|(y)\, dy \\ &\leq 2 \int_{I_n} |f f'|.\end{aligned}$$

Summing only the first N cells and applying Cauchy–Schwarz proves (2.7). The two expressions before subtraction on its left are nonnegative and increasing in N. Since their difference is uniformly bounded, one has a finite limit exactly when the other does. Taking these finite limits proves (2.8) and the domain identity. No subtraction of divergent series is used. □

Lemma 2.3 (Bounded exponential error)**.** *For all $x > 0$ away from the vertices,*

$$|V_\kappa(x) - C e^{x/\pi}| \leq B. \tag{2.9}$$

In particular (2.2) *holds.*

Proof. Integral comparison for $1/x$ shows that $h_n - \log n$ decreases and $h_n - \log(n+1)$ increases to the same limit γ. Their difference is $\log(1 + 1/n) \to 0$. Thus

$$\log n + \gamma < h_n < \log(n+1) + \gamma, \qquad n \geq 1. \tag{2.10}$$

The same comparison gives $0 < \gamma < 1$; for positivity one may use $h_n - \log(n+1) \geq 1 - \log 2 > 0$. For $x \in I_n$ with $n \geq 2$, (2.10) gives $n - 1 < e^{x/\pi - \gamma} < n + 1$. For the first cell the lower bound is replaced by $0 < e^{-\gamma} \leq e^{x/\pi - \gamma}$, and the upper bound remains two. Hence $|n - e^{x/\pi-\gamma}| \leq 1$ on every cell, which proves (2.9). An additive bounded difference between the two nonnegative weights does not change their weighted L^2 domains after intersection with unweighted L^2. Lemma 2.2 now proves (2.2). □

For $a, C > 0$ introduce the form

$$Q_{a,C}[f] = a \, \|f'\|_2^2 + \int_0^\infty C e^{x/\pi} |f(x)|^2 \, dx, \qquad f \in \mathcal{D}, \tag{2.11}$$

and its associated operator $A_{a,C}$. Young's inequality in (2.8) gives, for $0 < \varepsilon < 1$,

$$Q_{1-\varepsilon,C}[f] - d_\varepsilon \, \|f\|_2^2 \leq q_\kappa[f] \leq Q_{1+\varepsilon,C}[f] + d_\varepsilon \, \|f\|_2^2, \qquad d_\varepsilon = \frac{\kappa^2}{\varepsilon} + B. \tag{2.12}$$

The exact coefficient $C = \kappa e^{-\gamma}/\pi$ matters: replacing it by an unspecified comparable constant would lose the linear term of (2.3).

2.2 Closedness, core and operator domain

The form norm $(\|f\|_2^2 + Q_{1,C}[f])^{1/2}$ is complete. A Cauchy sequence converges in H_0^1 and in the weighted L^2 space, and the two limits agree in unweighted L^2. At $\varepsilon = 1/2$, (2.12) shows equivalence of this norm and $(\|f\|_2^2 + q_\kappa[f])^{1/2}$. Indeed its lower bound controls $Q_{1/2,C}$ by $q_\kappa + (2\kappa^2 + B)\,\|f\|_2^2$, while $Q_{1,C} \le 2Q_{1/2,C}$; the upper bound gives the reverse norm estimate. Therefore q_κ is closed.

The space $C_c^\infty(0,\infty)$ is a form core. Multiply $f \in \mathcal{D}$ by a smooth cutoff equal to one on $[0,R]$, zero beyond $R+1$, and with uniformly bounded derivative. The tails of $|f|^2$, $|f'|^2$ and $e^{x/\pi}|f|^2$ prove convergence in the weighted Sobolev norm as $R \to \infty$. For a fixed compactly supported cutoff the weight is bounded above and below on a containing finite interval. The usual H_0^1 smooth approximation on that interval then converges in the same weighted norm. This proves the core statement; it also proves dense definition in $\mathcal{H}$. The representation theorem for closed nonnegative forms yields a unique self-adjoint H_κ.

Its domain can be written without formal delta symbols:

$$\begin{aligned} D(H_\kappa) = \{f \in \mathcal{D} : f|_{I_n} \in H^2(I_n) \quad (n \ge 1),\ \sum_n \|f''\|^2_{L^2(I_n)} < \infty, \\ f'(x_n+) - f'(x_n-) = \kappa f(x_n) \quad (n \ge 1)\}, \\ (H_\kappa f)|_{I_n} = -f''|_{I_n}. \end{aligned} \tag{2.13}$$

To derive this description from the weak form, test within one cell to obtain $-f'' = g \in L^2$, hence H^2 regularity there. A test crossing one vertex yields the displayed derivative jump. Conversely, the local identities in (2.13) give $q_\kappa(f,v) = \langle g, v\rangle$ for compactly supported smooth v. At x_n integration by parts contributes $f'(x_n-) - f'(x_n+)$, cancelling $+\kappa f(x_n)$. The form-core approximation extends the identity to every $v \in \mathcal{D}$. It follows that f belongs to the associated operator domain. This argument does not require an unproved boundary term at infinity.

2.3 Compactness, positivity and simplicity

A form-norm bounded sequence has bounded H^1 norm and bounded $\int Ce^{x/\pi}|f|^2$ by (2.12). Its tails obey

$$\int_R^\infty |f|^2 \le C^{-1} e^{-R/\pi} \int_R^\infty Ce^{x/\pi}|f|^2. \tag{2.14}$$

Local Rellich compactness on $[0,R]$, followed by a diagonal subsequence and the uniform tail estimate, gives convergence in $\mathcal{H}$. Thus the form embedding, and hence the resolvent, is compact. The compact spectral theorem supplies the discrete eigenvalue list and a complete orthonormal basis. There are infinitely many eigenvalues because $\mathcal{H}$ is infinite-dimensional and the resolvent is injective.

If $H_\kappa f = 0$, then (2.1) gives $f' = 0$ almost everywhere. The only such L^2 function is zero. Zero cannot be a finite accumulation point of a compact-resolvent spectrum, so $E_1(\kappa) > 0$. For a fixed eigenvalue E, an eigenfunction solves $f'' + Ef = 0$ on the first cell and has $f(0) = 0$. Its initial derivative determines it there. Continuity and the jump in (2.13) propagate the two Cauchy data through all later cells. Zero initial derivative gives the zero function, so the eigenspace is at most one-dimensional. Self-adjointness excludes generalized eigenvectors. This proves part 1 of Theorem 2.1, including simplicity at every finite positive coupling.

3 A uniform exponential comparison

The coefficient of the kinetic term in (2.12) will depend on energy. For that reason a fixed-parameter exponential-potential asymptotic is not enough. The following estimate controls the whole compact parameter interval needed later.

Lemma 3.1 (Uniform comparator count)**.** *For each fixed $C > 0$ there exist $r_0(C) > 0$ and $M_C < \infty$ such that, for every $a \in [1/2, 3/2]$ and real $r \geq r_0(C)$,*

$$\left| N_{A_{a,C}}(r^2) - \frac{2r}{\sqrt{a}} \left[\log \frac{2r}{\sqrt{C}} - 1 \right] \right| \leq M_C. \tag{3.1}$$

Proof. Under the unitary rescaling $x = 2\pi y$, the operator becomes

$$A_{a,C} \simeq \frac{a}{4\pi^2} \mathcal{B}_b, \qquad \mathcal{B}_b = -\frac{d^2}{dy^2} + b^2 e^{2y}, \qquad b = 2\pi \sqrt{C/a}. \tag{3.2}$$

It has a Dirichlet condition at $y = 0$. For energy r^2 define $K = 2\pi r/\sqrt{a}$. The parameter b ranges over one fixed compact interval $[b_-, b_+] \subset (0, \infty)$, depending only on C. The weighted-domain and compactness argument of Section 2 gives a positive compact-resolvent operator $\mathcal{B}_b$.

The change $z = be^y$ turns its eigenvalue equation at energy K^2 into the modified Bessel equation of order iK. The decaying solution is $K_{iK}(be^y)$. For each fixed K its large-y asymptotic is $\sqrt{\pi/(2be^y)}e^{-be^y}$, whereas the independent growing solution has size $e^{be^y}/\sqrt{2\pi be^y}$. These are the classical modified-Bessel solution asymptotics [5, 10.25.3 and 10.40.1]. Thus the square-integrable solution at infinity is unique up to a scalar, and the positive eigenvalues are characterized exactly by

$$K_{iK}(b) = 0.$$

Here the subscript denotes imaginary *order*, not an imaginary spatial argument.

Put $d = b^2/4$. The Bessel series and connection identity [5, 10.25.2 and 10.27.4] give, for $K > 0$,

$$\begin{aligned} I_{-iK}(b) &= \frac{(b/2)^{-iK}}{\Gamma(1 - iK)} S_b(K), \\ S_b(K) &= \sum_{j=0}^{\infty} \frac{d^j}{j!(1 - iK)_j}, \\ K_{iK}(b) &= \frac{\pi \operatorname{Im} I_{-iK}(b)}{\sinh(\pi K)}. \end{aligned} \tag{3.3}$$

The last sign follows directly from $\sin(\pi i K) = i \sinh(\pi K)$ and $I_{iK}(b) = \overline{I_{-iK}(b)}$. The rising product $(1 - iK)_j$ has modulus at least K^j. Differentiation of its reciprocal introduces at most j factors, each of modulus at most $1/K$. Normal convergence of the series and its derivative on $K \geq K_0 > 0$ justifies termwise differentiation. Writing $d_+ = b_+^2/4$ therefore yields, for $K \geq 1$,

$$|S_b(K) - 1| \leq e^{d_+/K} - 1, \qquad |S_b'(K)| \leq \frac{d_+}{K^2} e^{d_+/K}, \tag{3.4}$$

uniformly in $b \in [b_-, b_+]$.

Choose one K_0 with $e^{d_+/K_0} - 1 < 1/2$. For $K \geq K_0$, all $S_b(K)$ then lie in the disk $|z - 1| < 1/2$, which carries a single logarithm. Its imaginary part is $O_C(K^{-1})$, and its derivative is $O_C(K^{-2})$. For the Gamma function use the logarithm analytic in the right half-plane and real at 1. The sectorial Stirling and digamma expansions [5, 5.11.1–2] imply

$$\begin{aligned}\operatorname{Im}\log\Gamma(1+iK) &= K\log K - K + \frac{\pi}{4} + O(K^{-1}),\\ \operatorname{Re}\psi(1+iK) &= \log K + O(K^{-2}).\end{aligned} \tag{3.5}$$

The sector is fixed and no b occurs in these estimates.

The unwrapped phase

$$\Psi_b(K) = K\log(2/b) + \operatorname{Im}\log\Gamma(1+iK) + \operatorname{Im}\log S_b(K)$$

is therefore well defined for $K \geq K_0$ and satisfies

$$\begin{aligned}\Psi_b(K) &= K\log(2K/b) - K + \frac{\pi}{4} + O_C(K^{-1}),\\ \Psi_b'(K) &= \log(2K/b) + O_C(K^{-2}) > 0\end{aligned} \tag{3.6}$$

after increasing the same K_0 if necessary. All bounds and this choice are uniform in the compact b interval. The amplitude in (3.3) is nonzero. Consequently, above K_0, the roots occur exactly once at each crossing of an integer multiple of π by Ψ_b. At a root, the inclusive convention changes the count relative to either neighboring open interval by at most one.

It remains to control the low-energy counting offset uniformly. Form order gives $\mathcal{B}_b \geq \mathcal{B}_{b_-}$. Hence the number of roots at or below K_0 is bounded by $N_{\mathcal{B}_{b_-}}(K_0^2) < \infty$. The phase $\Psi_b(K_0)$ is bounded on the same compact b interval. Thus both the low-energy count and the integer offset in the phase count are uniformly bounded. We obtain

$$N_{\mathcal{B}_b}(K^2) = \frac{K}{\pi}\left[\log(2K/b) - 1\right] + O_C(1). \tag{3.7}$$

Finally $K = 2\pi r/\sqrt{a}$ and $2K/b = 2r/\sqrt{C}$. Substitution into (3.7) proves (3.1). □

The use of special functions here evaluates a comparator, not the spectrum or determinant of the actual delta-comb. The estimates (3.4) and the uniform low-energy offset are what allow the kinetic coefficient a to move with the counting energy.

4 Counting law and spectral consequences

4.1 Transfer of the comparator estimate

We now prove part 2 of Theorem 2.1. For $k \geq 2$, choose $\varepsilon = 1/k$ in (2.12) and set $d_k = \kappa^2 k + B$. The min–max principle gives

$$N_{A_{1+1/k,C}}(k^2 - d_k) \leq N_\kappa(k^2) \leq N_{A_{1-1/k,C}}(k^2 + d_k). \tag{4.1}$$

For example, the upper form bound $q_\kappa \leq Q_{1+1/k,C} + d_k \|\cdot\|_2^2$ implies that every comparator eigenvalue at most $k^2 - d_k$ forces an eigenvalue of H_κ at most k^2, which establishes the left inequality.

For sufficiently large k depending on κ, both shifted energies are positive, and

$$r_\pm := \sqrt{k^2 \pm d_k} = k + O_\kappa(1).$$

Define

$$F(a,r) = \frac{2r}{\sqrt{a}}\left[\log(2r/\sqrt{C}) - 1\right].$$

For $a \in [1/2, 3/2]$ and $r = k + O_\kappa(1)$,

$$\partial_r F(a,r) = \frac{2}{\sqrt{a}} \log(2r/\sqrt{C}) = O_\kappa(\log k),$$
$$\partial_a F(a,r) = -\frac{r}{a^{3/2}} \left[\log(2r/\sqrt{C}) - 1\right] = O_\kappa(k \log k). \tag{4.2}$$

The mean value theorem and Lemma 3.1 therefore put both sides of (4.1) within $O_\kappa(\log k)$ of $F(1,k)$. Since

$$F(1,k) = 2k \log k + \left[\log(4/C) - 2\right] k = 2k \log k + C_\kappa k,$$

this proves (2.3). In particular, there exist $k_0(\kappa) > 1$ and $M(\kappa) < \infty$ such that the absolute error is at most $M(\kappa) \log k$ for every real $k \geq k_0(\kappa)$. The uniform comparator estimate already includes thresholds, so no exceptional sequence of energies is omitted.

4.2 Heat trace

In energy variables, the count is

$$N_\kappa(E) = E^{1/2} \log E + C_\kappa E^{1/2} + R_\kappa(E), \qquad R_\kappa(E) = O_\kappa(\log(E+2)) \quad (E \geq 1). \tag{4.3}$$

This estimate ensures summability of $e^{-tE_j(\kappa)}$ for every $t > 0$. Because the count vanishes below the positive ground energy and grows slower than an exponential at infinity, Stieltjes integration by parts gives

$$\operatorname{Tr}(e^{-tH_\kappa}) = t \int_0^\infty e^{-tE} N_\kappa(E)\, dE. \tag{4.4}$$

Extend R_κ to $(0,1)$ by (4.3). It is integrable there: the count is bounded and $E^{1/2} \log E$ is integrable. Its contribution to (4.4) on $(0,1)$ is $O_\kappa(t)$. For the high-energy part, put $u = tE$ and use

$$\log(2 + u/t) \leq \log(1/t) + \log(2+u) \qquad (0 < t \leq 1/2).$$

The error term in (4.4) is therefore $O_\kappa(\log(1/t))$. Differentiating the absolutely convergent Gamma integral at $3/2$ evaluates the two leading terms as

$$\frac{\Gamma(3/2)}{\sqrt{t}} \left[\log(1/t) + \psi(3/2) + C_\kappa\right].$$

The identities $\Gamma(3/2) = \sqrt{\pi}/2$ and $\psi(3/2) = 2 - \gamma - 2\log 2$ give $\psi(3/2) + C_\kappa = \log(\pi/\kappa)$. This proves (2.4) and part 3 of Theorem 2.1.

4.3 Source zeta continuation

For $\operatorname{Re} s > 1/2$, absolute convergence permits Stieltjes integration by parts:

$$Z_\kappa(s) = s \int_0^\infty N_\kappa(E) E^{-s-1}\, dE.$$

After splitting the integral at 1 and using (4.3), the result is

$$Z_\kappa(s) = \frac{s}{(s-1/2)^2} + \frac{sC_\kappa}{s-1/2}$$
$$+ s \int_1^\infty R_\kappa(E) E^{-s-1}\, dE + s \int_0^1 N_\kappa(E) E^{-s-1}\, dE. \tag{4.5}$$

The last integral is entire because the integrand vanishes below the positive ground energy. The preceding integral is holomorphic for $\operatorname{Re} s > 0$. Indeed, on a compact subset of that half-plane its integrand and each derivative with respect to s are bounded by an integrable multiple of $E^{-1-\eta}(1+\log E)^m$ for a suitable $\eta > 0$ and finite m. Equation (4.5) thus gives the claimed meromorphic continuation. Expansion of its first two terms at $s = 1/2$ gives (2.5). The double-pole coefficient is nonzero, so the pole is genuine and is the only one in $\operatorname{Re} s > 0$. This proves part 4 of Theorem 2.1. The remainder estimate does not extend this argument to $s = 0$.

4.4 Schatten threshold and ordinary determinant

For real $p > 0$, positivity gives

$$\sum_j E_j(\kappa)^{-p} = p \int_0^\infty N_\kappa(E) E^{-p-1}\, dE, \tag{4.6}$$

where both sides may be infinite. In (4.3), the positive leading term dominates all lower-order terms. Convergence in (4.6) is equivalent to convergence of

$$\int_1^\infty E^{-p-1/2} \log E\, dE,$$

which holds precisely for $p > 1/2$. At $p = 1/2$ the integrand is $(\log E)/E$ and diverges. The singular values of H_κ^{-1} are $E_j(\kappa)^{-1}$. For a fixed $z \in \rho(H_\kappa)$, $|E_j(\kappa) - z|/E_j(\kappa) \to 1$, so every such resolvent has the identical Schatten threshold. This proves part 5 of Theorem 2.1.

In particular the inverse is trace class. Its ordinary Fredholm determinant is the locally uniformly convergent entire product

$$D_\kappa(z) = \det(I - zH_\kappa^{-1}) = \prod_{j=1}^\infty \left(1 - \frac{z}{E_j(\kappa)}\right). \tag{4.7}$$

The zeros are exactly the simple source eigenvalues. Equation (4.7) is not a closed-form special-function evaluation of the comb determinant. The Bessel function in Section 3 belongs to $A_{a,C}$, not to H_κ. Nor does (4.5) justify a zeta-regularized value $Z_\kappa'(0)$.

5 The singular strong-coupling limit

Let H_∞ denote the orthogonal direct sum of the Dirichlet Laplacians on the intervals I_n. The strong-resolvent endpoint and its divisor spectrum are classical for this chain [4]. We include their form description to prove norm convergence and make the threshold convention explicit.

Theorem 5.1 (Dirichlet endpoint and unequal iterated limits)**.** *As $\kappa \uparrow \infty$, H_κ converges to H_∞ in norm resolvent sense. If eigenvalues are ordered with multiplicity, then*

$$E_j(\kappa) \uparrow E_j(\infty) \quad \textit{for every fixed } j. \tag{5.1}$$

For every fixed real E,

$$N_\kappa(E) = N_\infty(E) \quad \textit{for all sufficiently large } \kappa. \tag{5.2}$$

This includes energies that are eigenvalues of H_∞. At the endpoint,

$$\begin{aligned} N_\infty(k^2) &= \sum_{1\le n\le k} \lfloor k/n \rfloor = k\log k + (2\gamma - 1)k + O(\sqrt{k}), \\ Z_\infty(s) &= \zeta(2s)^2 \qquad (\operatorname{Re} s > 1/2). \end{aligned} \tag{5.3}$$

Consequently,

$$\begin{aligned}\lim_{\kappa\to\infty}\lim_{k\to\infty}\frac{N_\kappa(k^2)}{k\log k} &= 2,\\ \lim_{k\to\infty}\lim_{\kappa\to\infty}\frac{N_\kappa(k^2)}{k\log k} &= 1.\end{aligned} \tag{5.4}$$

5.1 Increasing forms and norm convergence

Fix $\kappa_0 > 0$. The forms q_κ for $\kappa \geq \kappa_0$ increase, and their supremum is finite exactly on

$$\mathcal{D}_\infty = \{f \in H_0^1(0,\infty) : f(x_n) = 0 \text{ for every } n \geq 1\}, \qquad q_\infty[f] = \|f'\|_2^2. \tag{5.5}$$

Every function in $\mathcal{D}_\infty$ belongs to $\mathcal{D}_\delta$: its sampled sum is zero. Lemma 2.2 then also places it in the weighted domain $\mathcal{D}$. Continuity of each finite-position trace shows that $\mathcal{D}_\infty$ is closed in H_0^1, so the limit form is closed. It is densely defined in $\mathcal{H}$ because finite sums of smooth functions supported in separate open cells are dense in $\bigoplus_n L^2(I_n)$.

The monotone convergence theorem for nonnegative closed forms therefore gives strong-resolvent convergence to the operator represented by (5.5). One may apply the theorem along an increasing integer sequence of couplings; form order squeezes intermediate couplings and gives the same continuous-parameter limit. Restriction to each cell identifies the limit form with the direct sum of the Dirichlet interval forms. Conversely, gluing cell functions whose endpoint traces are zero and whose L^2 and derivative norms are square summable gives a global H_0^1 function. The limiting operator is thus H_∞.

Set

$$R_\kappa = (H_\kappa + 1)^{-1}, \qquad R_\infty = (H_\infty + 1)^{-1}.$$

Resolvent order for nonnegative closed forms gives

$$0 \leq R_\kappa - R_\infty \leq R_{\kappa_0}, \qquad R_\kappa - R_\infty \longrightarrow 0 \quad \text{strongly.} \tag{5.6}$$

The upper operator R_{κ_0} is compact by Theorem 2.1.

Here compact domination upgrades strong to norm convergence. To see this directly, suppose $0 \leq T_\kappa \leq K$ for a compact positive operator K and that $T_\kappa \to 0$ strongly. Choose a finite-rank spectral projection P of K such that $\|(I-P)K(I-P)\| < \delta$. Then $\|(I-P)T_\kappa(I-P)\| < \delta$. Cauchy–Schwarz for the positive form $\langle T_\kappa \cdot, \cdot\rangle$ bounds each off-diagonal block by $\sqrt{\|K\|\,\delta}$. On the finite-dimensional range of P, strong convergence gives $\|PT_\kappa P\| \to 0$. The four-block decomposition now shows

$$\limsup_{\kappa\to\infty}\|T_\kappa\| \leq \delta + 2\sqrt{\|K\|\,\delta}.$$

Letting $\delta \downarrow 0$ proves norm convergence. Applying this argument to (5.6) gives $\|R_\kappa - R_\infty\| \to 0$, which is norm-resolvent convergence. This is an application of a general compact-operator principle, not a new abstract convergence theorem.

5.2 Eigenvalues and inclusive thresholds

The Dirichlet eigenvalues on I_n are

$$\left(\frac{\pi m}{\ell_n}\right)^2 = m^2 n^2, \qquad m \geq 1.$$

Thus j^2 has multiplicity $d(j)$, where $d(j)$ is the number of positive divisors of j. Only finitely many pairs (m,n) lie below a fixed energy, so this description also proves compactness of

R_∞. Norm convergence of compact positive resolvents and their min–max characterization imply convergence of each ordered resolvent eigenvalue, including repetitions. Transforming by $r \mapsto r^{-1} - 1$ proves convergence of each $E_j(\kappa)$. Form monotonicity and restriction of the form domain give

$$E_j(\kappa) \le E_j(\kappa') \le E_j(\infty) \qquad (\kappa \le \kappa'),$$

which completes (5.1).

Fix a real energy E and put $J = N_\infty(E)$. All indices $j \le J$ satisfy $E_j(\kappa) \le E_j(\infty) \le E$ and are counted at every finite coupling. The next endpoint eigenvalue satisfies $E_{J+1}(\infty) > E$. By (5.1), $E_{J+1}(\kappa) > E$ for sufficiently large κ, and then all later eigenvalues are excluded as well. This proves (5.2), including $E = j^2$. When $J = 0$, only the second part of the argument is needed.

The convention $\le$ is essential to this threshold statement: it uses convergence from below. For strict counts defined with $<$, convergence away from endpoint eigenvalues still follows, but equality at an endpoint eigenvalue is not asserted. The coupling threshold in (5.2) need not be uniform as $E \to \infty$.

5.3 Divisor count and limiting orders

Counting the pairs with $m^2n^2 \le k^2$ proves the first identity in (5.3). For real $k \to \infty$, let $L = \lfloor \sqrt{k} \rfloor$. Every pair $mn \le k$ has $m \le L$ or $n \le L$. The L^2 pairs with both inequalities have been counted twice, so

$$\sum_{n \le k} \lfloor k/n \rfloor = 2 \sum_{n \le L} \lfloor k/n \rfloor - L^2 = 2kh_L - L^2 + O(L). \tag{5.7}$$

The bounds (2.10) give $h_L = \log L + \gamma + O(1/L)$, since their difference is at most $1/L$. Using $L = \sqrt{k} + O(1)$ in (5.7) yields $k \log k + (2\gamma - 1)k + O(\sqrt{k})$. For $\operatorname{Re} s > 1/2$, absolute convergence gives

$$Z_\infty(s) = \sum_{m,n \ge 1} (m^2 n^2)^{-s} = \left(\sum_{n \ge 1} n^{-2s} \right)^2 = \zeta(2s)^2.$$

Finally (2.3) gives the first iterated limit in (5.4). At every fixed $k > 1$, (5.2) gives the inner limit in the second expression, and the endpoint asymptotic then gives one. This completes the proof of Theorem 5.1.

6 Finite numerical checks

The proofs above do not depend on numerical eigenvalues. This section reports finite checks of the constants and of two different numerical descriptions of the same finite head of the chain. All floating computations are non-certified: no interval enclosure of an infinite-spectrum count is claimed. The actual output, including the discrepancy on coarse grids, is retained with the source code.

Algebraic and sampling checks. Three SymPy identities vanish exactly: the linear coefficient after inserting $C = \kappa e^{-\gamma}/\pi$, the heat-trace constant, and the zeta principal part. At 70 decimal digits, the first 1,000 pairs of cell endpoints satisfy the tested bound in Lemma 2.3; the largest observed deviation of $e^{x/\pi - \gamma}$ from the cell index is approximately 0.5262051116, below one. For $f(x) = xe^{-0.4x}$, $\kappa = 1.5$, and 1,200 cells, the sampled-versus-cell-average difference is approximately 0.2534272797, against the global right-hand side 4.6875 in (2.7).

κ	k	Cells	c_{num}	$F_\kappa(k)$	$c_{\mathrm{num}} - F_\kappa(k)$
0.5	10	2545	64	64.065573	−0.065573
0.5	20	10085	156	155.857033	0.142967
0.5	40	40244	367	367.165840	−0.165840
1	10	1282	57	57.134101	−0.134101
1	20	5052	142	141.994089	0.005911
1	40	20132	339	339.439953	−0.439953
2	10	660	50	50.202629	−0.202629
2	20	2545	128	128.131146	−0.131146
2	40	10085	312	311.714066	0.285934

Table 1: Finite, non-interval-certified counts compared with $F_\kappa(k) = 2k \log k + C_\kappa k$. The integer c_{num} is the common shooting and two-finest-grid head count, not a rigorous certificate of $N_\kappa(k^2)$. Coefficients are the proved analytic values, not fitted to this table.

The integrals were computed from an explicit antiderivative rather than from the sampled values. These checks cover the stated finite endpoints and one test function, not all cells or functions. The divisor hyperbola identity was also checked exactly at integer frequencies $1, 2, 10, 100, 1000$, giving counts $1, 3, 27, 482, 7069$.

Independent finite-head descriptions. The shooting calculation propagates free solutions and delta jumps using a rounded Prüfer phase. The second calculation uses piecewise-linear finite elements, a consistent mass matrix, and negative LDL pivots of the stiffness matrix minus k^2 times the mass matrix. It uses neither transfer matrices nor trigonometry. Both methods compute Dirichlet and Robin counts for a finite head; the finite-element inertia computes strict counts. The selected test energies are numerically stable but have not been separated from eigenvalues by interval arithmetic.

For each pair (κ, k), the cutoff is

$$N = \left\lceil \frac{4\pi(k^2 + \kappa^2 + 1)}{\kappa} \right\rceil .$$

Splitting at x_N and assigning the cut-vertex penalty to the head gives a Robin head and a free-left-end tail as a lower form comparison; imposing a zero at the cut gives the Dirichlet comparison. The cell proof of Lemma 2.2 also works on this tail without a left Dirichlet condition. Young's inequality with $\varepsilon = 1/2$ bounds the tail form below by

$$\tfrac{1}{2} \|f'\|_2^2 + \left(\frac{\kappa(N+1)}{\pi} - 2\kappa^2 \right) \|f\|_2^2 .$$

The coefficient of $\|f\|_2^2$ exceeds k^2 at every selected cutoff, with computed ratio between 4.0029 and 4.1281. This explains the cutoff choice; it does not certify the rounded head counts.

For all nine cases in Table 1, the original and doubled shooting cutoffs agree, as do the two finest finite-element grids with maximum free phase 0.02 and 0.01. At these levels the head Dirichlet and Robin counts also agree. The two coarser grids have maximum free phase 0.08 and 0.04. At $\kappa = 2$, $k = 40$, both coarser grids count 311, while shooting and the finer grids count 312. Thus eight, not nine, cases agree at all four mesh levels. The smallest reported relative LDL pivot at the finest level is approximately 7.84×10^{-6}; it is a diagnostic, not an accumulated-roundoff bound or spectral-gap certificate.

The residuals in the table have absolute value below 0.44, but nine points cannot establish the asymptotic error, its implied constant, or an $O(1)$ refinement. The calculation used Python 3.12.3, SymPy 1.14.0, and mpmath 1.3.0 on a Linux x86_64 CPU environment. No target-zero data or coefficient fitting was used. The accompanying `sanity.py`, `SANITY_OUTPUT.json`, and `CHECK_REPORT.md` give the commands and complete output.

7 Discussion and limitations

The finite-coupling harmonic chain is controlled by an exponential confining potential at high energy. The sampling bound identifies that potential on the common closed-form domain, and the uniform phase estimate keeps the energy-dependent comparison error logarithmic. This proves the coefficient two in the finite-coupling count and its explicit linear correction. The Dirichlet limit instead has coefficient one. Norm-resolvent convergence, ordered eigenvalue convergence and inclusive threshold stabilization explain precisely how both descriptions can hold.

All finite-coupling estimates fix $\kappa > 0$ before taking the high-energy limit. No uniform remainder as $\kappa \to 0$ or $\kappa \to \infty$ is proved. At $\kappa = 0$, the closed physical form is the free Dirichlet half-line form on H_0^1, with spectrum $[0, \infty)$; substituting zero into the positive-coupling domain or asymptotic formulas is invalid. The singular change in the leading coefficient also prevents transferring the endpoint arithmetic to finite coupling by a formal substitution.

The source zeta continuation stops at $\operatorname{Re} s > 0$. The ordinary determinant exists by inverse trace class, but neither a value of $Z_\kappa'(0)$ nor an explicit Bessel formula for the comb determinant has been obtained. A finite-coupling orbit trace formula, a sharper remainder, and uniform asymptotics in a joint energy–coupling regime are separate questions not answered here.

Finally, an averaged spectral count is not a construction of a prescribed arithmetic zero set. The endpoint identity $Z_\infty(s) = \zeta(2s)^2$ is a source divisor identity; it gives no finite-coupling prime-orbit correspondence, target Euler factors, functional equation, or root number. The logarithmic remainder proved here also cannot be used as though it were a bounded remainder in an argument excluding a proposed full divisor match. No Hilbert–Pólya realization, or general impossibility theorem for such a realization, is claimed.

Draft status. This is an unnumbered mathematical manuscript. The complete proof is included above, and the numerical checks are explicitly separated from that proof. The original model and classical endpoint results remain attributed to their published owners. The bounded source audit accompanying the manuscript does not establish literature-wide novelty or submission readiness.

References

[1] Patrizio Bifulco. *Contributions to the Diffusion on Graphs and Networks: Regularity of Heat Kernels and a Heat Content Formula, Faber–Krahn inequality for the Heat Content, p-Torsional Rigidity, Spectral Comparison Results for Schrödinger Operators*. PhD thesis, FernUniversität in Hagen, 2025. URL https://d-nb.info/1388406829/34. Submitted 25 June 2025.

[2] Patrizio Bifulco and Joachim Kerner. Some spectral comparison results on infinite quantum graphs. *Journal of Mathematical Physics*, 65(7):073502, 2024. doi: 10.1063/5.0178226. URL https://arxiv.org/abs/2308.16869.

[3] Sebastian Egger. *The solution of the "constant term problem" and the zeta-regularized determinant for quantum graphs*. PhD thesis, Universität Ulm, 2011. URL https://oparu.uni-ulm.de/handle/123456789/1978. Defended 16 February 2012; repository record migrated in 2016.

[4] Sebastian Egger né Endres and Frank Steiner. An exact trace formula and zeta functions for an infinite quantum graph with a non-standard Weyl asymptotics. *Journal of Physics A: Mathematical and Theoretical*, 44(18):185202, 2011. doi: 10.1088/1751-8113/44/18/185202. URL https://arxiv.org/abs/1104.1364.

[5] National Institute of Standards and Technology. NIST Digital Library of Mathematical Functions, 2026. URL https://dlmf.nist.gov/. Version 1.2.7, released 15 June 2026. Sections 5.11, 10.25, 10.27 and 10.40; accessed 5 September 2026.